



AIMS05-02-002

Issue 3

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AIMS
Airbus Material Specification

Glass fibre reinforced epoxy prepreg
120 or 220 Style Glass fabric/Epoxy resin
(105 g/m² fibre area mass)

180°C curing class
Material Specification

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1 Scope

This Material Specification defines the requirements of epoxy resins reinforced with 105g/m² glass fiber fabric (120 or 220 style), 180°C curing class for aerospace application.

The requirements specified in this Material Specification shall not be used as stress allowables.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for the manufacture of non-structural components (components whose design does not take into account any structural performance of the glass prepreg – galvanic corrosion protection for example).

Materials to be delivered according to the relevant ABS.

3 Normative references

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-02-000 Part 1	Airbus Material Specification Glass fiber reinforced thermosetting prepreg Technical Specification/General requirements
AIMS 05-02-000 Part 3	Airbus Material Specification Glass fiber reinforced thermosetting prepreg Fabric / Epoxy resin, 180°C curing class Technical specification/Qualification and batch release testing
AIMS 05-02-000 Part 100	Glass fibre reinforced epoxy prepreg Acceptance Testing and Storage Life Prolongation
ABS 5009	EP-prepreg, glass fiber fabric preimpregnated with epoxy resin
AITM 1-0003	AirbusTest Method Fibre reinforced plastics, Determination of the glass transition temperatures
AITM 3-0001	AirbusTest Method Fibre reinforced plastics, Analysis of thermoset systems by high performance liquid chromatography
AITM 3-0002	AirbusTest Method Analysis of non-metallic materials (uncured) by Differential scanning calorimetry
AITM 3-0003	AirbusTest Method Analysis of non-organic compounds by infrared spectroscopy
AITM 3-0004	AirbusTest Method Determination of gel time and viscosity
AITM 3-0008	AirbusTest Method Determination of the extent of cure by differential scanning calorimetry
AMS3824	Cloth, Glass, Finished, For Resin Laminates
AMS-C-9084	Cloth, Glass, Finished, For Resin Laminates
EN2155-12	Aerospace series – Test method for the determination of linear thermal expansion
EN2329	Aerospace series - Test methods for the determination of mass per unit area

EN2330	Aerospace series - Textile glass fibre preimpregnated - test method for the determination of volatile content
EN2331	Aerospace series - Textile glass fibre preimpregnated - Test methods for the determination of the resin an fibre content of woven textile glass, and mass of fibre per unit area
EN2332	Aerospace series - Textile glass fibre preimpregnated - Test methods for the determination of the resin flow
EN2377	Aerospace series – Glass fibre reinforced plastics – Test method. Determination of apparent interlaminar shear strength
EN2743	Aerospace series - Fibre reinforced plastics. standard procedures for conditioning prior to testing unaged materials
EN2823	Aerospace series - Fibre reinforced plastics - determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN3615	Aerospace series - Fibre reinforced plastics - procedure for the determination of the conditions of exposure to humid atmosphere and the determination of moisture absorption
ISO 1183 Method A	Plastics - method for determining the density and relative density of non - cellular plastics
ISO 1889	Reinforcement yarns - determination of the linear density

4 Technical requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material description

See Table 1.

Table 1: Material description

No.	Characteristic		Requirements
1	Material type		Glass fiber fabric / epoxy resin
2	Formulation	Fibre	Glass fiber, type E
		Fabric	120 or 220 style, 4H satin weave (AMS3824C and AMS-C-9084)
		Matrix	Modified epoxy resin
		Sizing	Epoxy compatible
3	Resin content by weight, uncured		as defined in table 4
4	Material code		as defined in table 4

4.1.2 Material Application

This specification defines the requirements for one type of material application: **non-structural**.

The different possible applications of glass fiber fabric reinforced epoxy prepregs are defined in the Technical Specification AIMS 05-02-000 Part 3.

4.1.3 Work life and storage life

See Table 2.

Specific requirements for storage conditions are listed in the relevant IPS.

Table 2: Worklife and storage conditions

No.	Properties	Shop condition	Symbol	Unit	Requirements
1	Storage life at -18°C	-	-	months	≥ 12
2	Worklife ¹⁾ (tack life)	30°C/75% R.H.	-	days	≥ 10
		(25+2)°C/(45+10)% R.H.	-	days	≥ 10
¹⁾ The prepreg can be exposed to an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use.					

4.1.4 Processing

The curing cycle shall be agreed between the manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave. The curing cycle shall comply with the general requirements reported in Table 3.

The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all the batch release test specimens.

Table 3: Processing conditions

No.	Property	Unit	Requirements
1	Heating rate	°C/min	0,2 - 5,0
2	Cure temperature	°C	180 +10/-5
3	Cure time	min	120 - 240
4	Cure pressure	MPa	0,3 – 1,1
5	Cooling rate	°C/min	0,2 - 3,5
6	Vacuum (abs)	MPa	0,005 - 0,1
Note: The use of an intermediate temperature and low pressure dwell is permissible if agreed between manufacturer and supplier, and specified in the QTP or proven suitable by subsequent tests			

4.2 Specific

4.2.1 Physical-chemical properties

Uncured and cured properties see Table 4.

Table 4: Physical-Chemical properties, all application

No.	Property		Test Method	Unit	Requirements ²⁾
Application					NS ¹⁾
Material					AIMS05-02-002 A
Uncured					
1	Prepreg areal weight		EN2329	g/m ²	(171 to 192) ⁶⁾ ± 15
2	Fibre areal weight ³⁾		EN2331	g/m ²	105 ± 5
3	Resin density		ISO 1183 A	g/cm ³	⁴⁾
4	Fibre density		ISO 1889	g/cm ³	2,56 ± 0,06
5	Volatile content ³⁾		EN2330	%	≤ 2
6	Resin content ³⁾		EN2331	%	(40 to 45) ⁶⁾ ± 3
7	Physico-chemical tests ⁵⁾	HPLC	AITM 3.0001A	-	⁴⁾
		DSC	AITM 3.0002	-	
		IR	AITM 3.0003	-	
		Gel time	AITM 3.0004 B or C	-	
		Viscosity	AITM 3.0004 A	-	
8	Resin flow		EN2332	%	⁴⁾
9	Tack		As defined in AIMS 05-02-000 Part 1	-	⁴⁾
Cured					
10	CTE ⁵⁾ (warp and weft)		EN2155-12	-	⁴⁾
11	Water uptake		EN3615 (70°C - 85% R.H. equiv.)	%	≤ 2,0
12	Extent of cure		AITM 3.0008	-	⁴⁾
Note: the stated requirements are average values					
¹⁾ Non-structural application					
²⁾ 3 specimens per batch					
³⁾ Method to be agreed before qualification between purchaser and supplier.					
⁴⁾ Equipment settings and value to be agreed between manufacturer and Airbus. Result to be reported.					
⁵⁾ CTE is Coefficient of thermal Expansion					
⁶⁾ Nominal to be agreed between purchaser and supplier before qualification.					

4.2.2 Mechanical properties

See Table 5.

Table 5: Mechanical properties, non-structural application

No.	Properties		Test method	Unit	Conditioning ⁴⁾	Test temp. [°C]	Requirements	
							X _{ave} ¹⁾	X _{min} ²⁾
Application							NS ³⁾	
Material							AIMS 05-02-002 A	
1	Glass transition temperature	T _{g onset}	AITM 1.0003	°C	dry	-	-	140
					wet	-	-	110
2	Resistance to aggressive fluids							
2a	Reference		EN2377	MPa	dry	RT	5)	5)
					wet	70	5)	5)
2b	Hydraulic fluid	70°C / 1000h			70	6)	6)	
2c	Hydraulic fluid / water	50°C / 1000h			70	6)	6)	
2d	Fuel	RT/1000h			70	6)	6)	
2e	MEK	RT/1h			RT	7)	7)	
2f	Resistance to paint strippers (Turco 1270-5)		AITM 1.0002 + ₈₎		dry	RT	7)	7)
1) Minimum average value 2) Minimum individual value 3) Non-structural application 4) Conditioning: “dry” according to EN 2743; “wet” is 70°C/85% r.h. according to EN 2823 5) To be reported 6) Values after conditioning will have no decrease compared to the reference value at (70°C/85%R.H.), tested at the same temperature. 7) Values after conditioning will have no decrease compared to the reference value at RT/dry 8) AIMS 05-02-000 Part 1, chapter 4.4.6								

5 Designation

Refer to relevant ABS5009.

6 Technical specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-02-000 Part 1 and Part 3.

7 Incoming inspection

Materials qualified against this Standard shall undergo "Technical Incoming Inspections" as specified through AIMS 05-02-000 Part 100 prior to use

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 11/00		New standard
2 05/06	all	complete revision, revision of the amount of tests required
3 11/14	Chapter 1 Table 1 Chapter 7 -	220 style added MIL-C-9084 Type III for fabric formulation is replaced by AMS-C-9084 and AMS3824C. Chapter 7 defined to link to Part 100 specification for Incoming Inspection Document updated to latest template